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A Participatory Approach to Support Glucose Tolerance Tracking in Real-Life of Patients With Type 2 Diabetes (GluToTrack)

ClinicalTrials.gov ID NCT06864546

Sponsor Enea Parimbelli

Information provided by Enea Parimbelli, University of Pavia (Responsible Party)

Last Update Posted 2025-04-30

Recruiting

Find contact information
(https://clinicaltrials.gov/study/NCT06864546#contact-and-locations)

Check who can join
(https://clinicaltrials.gov/study/NCT06864546#participation-criteria)

Researcher View Tab

Trial Contacts

Contacts *	ICMJE

1. Contact: Enea Parimbelli
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Study Record Dates

First Submitted	ICMJE
2025-03-03	

First Posted	ICMJE
2025-03-07	

Last Update Posted
2025-04-30

Last Verified *
2025-03

Outcome Measures

Change History

See all versions of this study

Primary (Current) * ICMJE

(Submitted: 2025-03-07)

- Change in Blood Glucose Control Over the Two Study Weeks [Time Frame: From enrollment to the end of the treatment at 2 weeks]
  - The estimation of the Disposition Index (DI) requires plasma measurements in hospitalized setting. The investigators propose a model able to estimate the DI from CGM data collected in outpatient condition.

    The DI (index of glucose tolerance) quantified using mathematical models from patient generated health data and CGM data, and the reference one, from plasma measurements, will be calculated to assess if these are affected or not by physical activity. Also the area under the glucose curve measured by CGM device will be evaluated to assess the intervention effect.

Primary (Original) * ICMJE

(Submitted: 2025-03-03)

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Secondary (Current) [*] ICMJE

Not provided

Secondary (Original) [*] ICMJE

Not provided

Other Pre-specified (Current)

Not provided

Other Pre-specified (Original)

Not provided

Trial Description

Brief Title * ICMJE

A Participatory Approach to Support Glucose Tolerance Tracking in Real-Life of Patients **With Type 2 Diabetes**

Official Title *§ ICMJE

A Participatory Approach to Support Glucose Tolerance Tracking in Real-Life of Patients **With Type 2 Diabetes**

Brief Summary *

Recent studies show that lifestyle interventions, such as exercise, healthy diet, and education, can help prevent type 2 diabetes in high-risk individuals, and even reverse the disease in its early stages. However, not many methods have been developed to use modern technology to help people manage their diabetes in a more active way, in line with participatory medicine, where patients play a key role in their treatment.

In this study, the investigators propose an e-health approach to automatically collect health data from patients, including information from continuous glucose monitor (CGM) and other health tracking devices, in real-life conditions. The investigators will also develop a simple and easy-to-understand tool to track patient's metabolic status and will analyze how it relates to lifestyle changes using the data collected during the study.

Twenty sedentary individuals with type 2 diabetes, not on insulin treatment, will take part in the trial. All patients will be monitored for two weeks using a CGM device and an activity tracker. During the first week, participants will follow their normal daily routine, while in the second week, subjects will be asked to engage in moderate physical activity every day, consisting of walking. At the end of each week, patients will take a meal tolerance test.

The main goal of the study is to measure how blood sugar levels change over the two weeks, also evaluating the impact of individuals' daily activities like exercise on it. The research team will develop and use mathematical models to measure this change. A secondary goal is to evaluate the ease of use of the e-health system for data collection.

Detailed Description

Diabetes Mellitus (DM) represents one of the major and fastest growing global health emergencies. Approximately 90% of the individuals with DM are affected by type 2 diabetes (T2D), whose management usually begins with lifestyle interventions, then progressing to one or more antihyperglycemic, while only ultimately moving to insulin therapy. Focusing on non-insulin treated T2D, evidence has shown lifestyle modifications, including exercise, healthy diet and education, to be effective in reversibility of the disease, but also in preventing the transition to T2D in high-risk (healthy) individuals. This would call for the improvement of participatory medicine, exploiting the increase in smartphone use over the past decade. In this framework, technological platforms would be key for the development of personalized, patient-centered interventions that integrate patient-reported data, tailored education, and individualized feedback. A fundamental component of these technological platforms for participatory medicine in diabetes is represented by continuous glucose monitoring (CGM) devices. These devices, usable in real-life conditions, provide prompt information on glucose trends and variability, warnings for out of-range glucose values as well as the ability for physicians to remotely review glucose profiles. Numerous studies have shown benefits of using CGM devices mainly in T1D but recently also in advanced T2D, while only limited data is available in non insulin treated T2D. The increasing accuracy of these devices paved the way for the development of tools assessing glucose tolerance in real-life conditions. The development of these tools would be highly beneficial for patients with T2D, allowing to provide information on therapy effectiveness and, possibly, disease progression. Moreover, incorporation of these tools into technological platforms, empowering participatory interventions and supporting self-management, would be fundamental for both the patients and the healthcare system.

The aim of the GluToTrack project is to combine bioengineering mathematical models and a data collection platform based on m-health solutions to develop a tool for glucose tolerance tracking, as well as evaluating the impact of individuals' daily activities like exercise on it, from PGHD and CGM data collected in real-life conditions. Data, required for the mathematical model validation phase, will be gathered, exploiting an e-health approach combining medical informatics and medical device wearables, running an in vivo clinical trial in a population of 20 patients with non-insulin treated T2D.

The clinical trial is an interrupted time series analysis. The trial seeks to enroll 20 participants with non-insulin treated T2D, 40 to 70 years old and physically inactive (less than 150 min/week of moderate physical activity).

During the first visit (visit 1, day 0) routine vital signs will be registered, and wearables (a CGM device and an activity tracker) and smartphones provided. The study is structured in two 1-week-long phases: in the first week patients will be asked to maintain their already sedentary lifestyle, while during the second week, participants should take at least 10,000 steps/day, with sitting replaced by standing and light-intensity walking, for a total of at least 150 min/week of light physical activity. At the end of the first week (visit 2, day 7) participants undergo an MMTT, which consists in the ingestion of a meal containing 75 g of carbohydrates and the drawing of 10 plasma samples for the measurement of plasma glucose, insulin and C-peptide concentrations. At the end of the second week (visit 3, day 14), subjects will repeat the MMTT to also assess the potential impact of the physical activity intervention on glucose tolerance.

Study Type * ICMJE

Interventional

Study Phase * ICMJE

Not Applicable

Study Design *§ ICMJE

Allocation
N/A

Interventional Model
Single Group Assignment

Masking
None (Open Label)

Primary Purpose
Basic Science

Condition * ICMJE

- **Type 2 Diabetes Mellitus, Non Insulin Dependent**

Intervention * ICMJE

- Behavioral: Physical activity intervention
    - In the first week, patients will be asked to maintain their normal sedentary life, spending most of the waking day sitting. During the second week, participants will be asked to take at least 10,000 steps/day, with sitting replaced by standing and light-intensity walking.

      See doi: 10.1007/s00125-016-4161-7

Study Arms * ICMJE

- Experimental: Physical activity intervention
  - Study participants will be monitored for two weeks with a CGM device and an activity tracker. During the first week, participants will follow their usual daily routine, while in the second week, subjects will be asked to engage in daily physical activity, consisting of walking
  - Interventions:
    - Behavioral: Physical activity intervention

Publications

(Includes general and study results' publications, and Pubmed publications referencing this study by ClinicalTrials.gov Identifier (NCT Number))

- Jackson MA, Ahmann A, Shah VN. Type 2 Diabetes and the Use of Real-Time Continuous Glucose Monitoring. Diabetes Technol Ther. 2021 Mar;23(S1):S27-S34. doi: 10.1089/dia.2021.0007. (https://pubmed.ncbi.nlm.nih.gov/33534631)

- Dehghani Zahedani A, Shariat Torbaghan S, Rahili S, Karlin K, Scilley D, Thakkar R, Saberi M, Hashemi N, Perelman D, Aghaeepour N, McLaughlin T, Snyder MP. Improvement in Glucose Regulation Using a Digital Tracker and Continuous Glucose Monitoring in Healthy Adults and Those with Type 2 Diabetes. Diabetes Ther. 2021 Jul;12(7):1871-1886. doi: 10.1007/s13300-021-01081-3. Epub 2021 May 28. (https://pubmed.ncbi.nlm.nih.gov/34047962)

- Duvivier BM, Schaper NC, Hesselink MK, van Kan L, Stienen N, Winkens B, Koster A, Savelberg HH. Breaking sitting with light activities vs structured exercise: a randomised crossover study demonstrating benefits for glycaemic control and insulin sensitivity in type 2 diabetes. Diabetologia. 2017 Mar;60(3):490-498. doi: 10.1007/s00125-016-4161-7. Epub 2016 Nov 30. (https://pubmed.ncbi.nlm.nih.gov/27904925)

Recruitment Information

Recruitment Status * ICMJE

Recruiting

Enrollment (Estimated) *§ ICMJE

(Submitted: 2025-03-03)

20

Original Enrollment (Estimated) *§ ICMJE

Same as current

Study Start Date (Estimated) *§ ICMJE

2025-05

Primary Completion Date (Estimated) *

2025-12 (Final data collection date for primary outcome measure)

Study Completion Date (Estimated) *§ ICMJE

2026-01-15

Eligibility Criteria * ICMJE

Inclusion Criteria:

- Non-insulin treated T2D diagnosis (not treated at all or only with metformin);
- Physically inactive (less than 150 minutes/week of moderate physical activity);
- Age between 40 and 70;
- HbA1c below 8.5%.

Exclusion Criteria:

- Diabetes diagnosis less than 3 years before the study initiation;
- Pregnancy;
- Symptomatic heart disease, e.g. history of myocardial infarction, coronary bypass, stenting procedure, angina, or any ischemic cerebrovascular event;
- Use of a medication that significantly impacts glucose metabolism (oral steroids);
- Use of a medication that significantly lowers heart rate (beta blockers, reserpine, guanethidine, methyldopa, clonidine, cimetidine, digitalis, calcium channel blockers, amiodarone, antiarrhythmic drugs, or lithium);
- Atrial fibrillation;
- Use of an electronic pacemaker.

Sex/Gender * ICMJE

Sexes Eligible for the Study:
All

Ages * ICMJE

40 Years to 70 Years (Adult, Older Adult)

Accepts Healthy Volunteers *§ ICMJE

No

Location Countries ICMJE

Italy

Removed Location Countries

Administrative Information

NCT Number ICMJE

NCT06864546

Other Study ID Numbers [*] ICMJE

CTSPV46-24

P2022XSFA7(Other Grant/Funding Number)

F53D23008460001(Other Grant/Funding Number)

C53D23007310001(Other Grant/Funding Number)

C53D23007310001(Other Grant/Funding Number)

Has Data Monitoring Committee

No

U.S. FDA-regulated Product *§

Studies a U.S. FDA-regulated Drug Product:
No

Studies a U.S. FDA-regulated Device Product:
No

Product manufactured in and exported from the U.S.:
No

IPD Sharing Statement ICMJE

Plan to Share IPD:
Undecided

Current Responsible Party *

Enea Parimbelli, University of Pavia

Original Responsible Party *

Same as current

Current Study Sponsor * ICMJE

Enea Parimbelli

Original Study Sponsor * ICMJE

Same as current

Collaborators ICMJE

Istituti Clinici Scientifici Maugeri SpA

University of Padova

Investigators ICMJE

Not provided

PRS Account

University of Pavia

Symbol Legend

*	Required
*§	Required if Study Start Date is on or after January 18, 2017
[*]	Conditionally required
No Symbol	Unmarked fields are optional. If no information is provided, fields will be labeled as "not provided"
ICMJE	Data element required by the International Committee of Medical Journal Editors (https://www.icmje.org/recommendations/browse/publishing-and-editorial-issues/clinical-trial-registration.html) and the World Health Organization ICTRP (https://www.who.int/clinical-trials-registry-platform)